



EMBEDEDD SYSTEM LAB

Project: Escape Room - Digital Fort

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Introduction

The “Escape Room – Digital Fort” project is a game-based control system implemented using the PIC16F877A microcontroller. The system simulates a locked room containing three sequential logic challenges that a player must solve to “escape.” Each challenge tests fundamental embedded system design principles, including digital I/O, ADC conversion, LCD/7-segment display interfacing, and logical decision-making—all written entirely in assembly language. The project was developed using MPLAB for programming and Proteus for simulation and testing.

System Description

The system is composed of three core subsystems, each responsible for one challenge. These subsystems interact with player input through switches and buttons, and provide feedback via LEDs, an LCD, and a 7-segment display.

Challenge 1: Prime Number Check

- The player inputs a 4-bit number using switches.
- The system checks if the number is prime by comparing it with a hardcoded list of small primes.
- If the number is prime, a green LED is turned ON and a value is shown on the 7-segment display.

Challenge 2: Range Detection via ADC

- The player rotates a potentiometer.
- The PIC reads the analog voltage using its internal ADC and determines the range it falls into.
- If the range matches a pre-defined correct range, the LCD displays “You are correct” and a green LED lights up.
- If it’s close (adjacent range), it shows “You are so close.” Otherwise, nothing is displayed.

Challenge 3: Math Challenge with Timer

- The player presses a Start button to trigger a random math problem on the LCD.
- A countdown starts from 9 and is displayed on the 7-segment display.
- The player must input the answer using switches and press a Finish button before the countdown ends.
- If the answer is correct, the system displays “YOU WIN! Escape” and activates all LEDs and a buzzer.

If the player fails any challenge, the system resets to Challenge 1 and displays “Hard Luck.”

Hardware System

Microcontroller:

- **PIC16F877A** is used due to its rich peripheral set including ADC, I/O ports, and ease of use in assembly.

Inputs:

- 4-bit switches for number input
- Pushbuttons (Start/Finish)
- Potentiometer for analog input (via RA0)

Outputs:

- 16x2 LCD Display (PORTD, controlled via PORTA pins)
- 7-Segment Display (PORTC)
- LEDs (connected to PORTE)

Initialization Details:

- All ports are set up in initial subroutine.
- ADCON1 and ADCON0 are configured for single-channel analog input (RA0).
- TRIS registers are set according to input/output use.
- LCD is initialized with standard 8-bit, 2-line mode.

System Testing and Results

Methodology:

- The system was tested using Proteus with interactive components (buttons, pot, etc.).
- Each challenge was tested individually and in sequence to ensure complete functionality.

Test Cases:

Challenge	Input	Expected Output	Observed Result
Prime Check	0b0101 (5)	LED ON, 7-Segment shows result	✓ Pass
Prime Check	0b0110 (6)	LED OFF, 7-Segment shows result	✓ Pass
Range Check	Potentiometer at 50%	“You are correct” on LCD, LED ON	✓ Pass
Range Check	Pot at 40% or 60%	“You are so close”	✓ Pass
Math Challenge	Random (e.g. 2+3)	Correct answer entered & timer active	✓ Win message shown
Math Challenge	Wrong answer / timeout	“Hard Luck” message	✓ Reset occurred

Discussion:

- Proper debouncing and delays were included to handle button press reliability.
- LCD and 7-segment interfacing worked consistently across Proteus simulations.
- ADC range detection was carefully calibrated using test voltages.

Conclusion

This project successfully demonstrates the design and implementation of a mini embedded game using only PIC assembly. From low-level ADC reading and display interfacing to system logic and player feedback, the project integrates various embedded system techniques.

Group Contributions:

Student Name	Contribution
mohammad	Prime number logic, 7-segment implementation
Marlo	ADC configuration, range checking logic
Heyam	LCD interfacing, Math challenge logic
izzeldeen	System integration, Proteus simulation

Challenges Encountered:

- Implementing accurate delay logic using nested loops
- Ensuring random problem generation stays within bounds
- Coordinating LCD and 7-segment outputs in parallel

Despite the challenges, the team achieved a fully functional and reliable design, fulfilling all project requirements.